



26-Port 10G Core Routing Switch (NAV-C24S2Q)

High-Speed Layer 3 Industrial Connectivity

The **NAV-C24S2Q** is a high-performance Layer 3 switch featuring 24x 1/10G SFP+ ports and 2x 40G QSFP uplink ports. Designed for next-generation enterprise networks and data centers, it delivers line-speed switching, advanced QoS, and robust security.



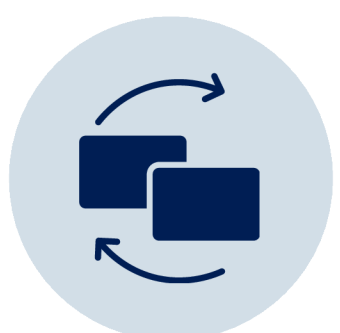
High-Speed Architecture

Features a non-blocking switching capacity of 2.56Tbps and a forwarding rate of 476.16Mpps for seamless big data transmission.



Advanced L3 Routing

Supports complete IPv4/IPv6 routing protocols including RIP, OSPF, and BGP to meet complex network interconnection needs.



Robust Security

Includes IEEE 802.1x authentication, ACL filtering, and hardware-level protection against DoS attacks and broadcast storms

Switching Capacity

2.56Tbps

High-performance core layer handling

Forwarding Rate

476 Mpps

Wire-speed forwarding @64byte

This device provides a rich server access solution with comprehensive reliability features. It supports VRRP for redundancy, intelligent fan speed regulation for energy efficiency, and diverse management options (Web/CLI/SNMP). Ideal for smart campuses, telecom aggregation, and large-scale video surveillance networks.

26-Port 10G Core Routing Switch (NAV-C24S2Q)



Product Overview

The **NAV-C24S2Q** is a high-performance Layer 3 routing switch designed for 10G access and 40G uplink applications. It is engineered to meet the demands of next-generation enterprise networks, data centers, and metropolitan area networks (MAN).

This device serves as an ideal solution for the core layer of small-to-medium enterprises or the aggregation layer of campus networks and telecom operators. The NAV-C24S2Q features 24x 1/10G SFP+ fiber ports and 2x 40G QSFP fiber uplink ports in a standard 1U/19-inch rack-mount form factor.

With a massive switching capacity of 2.56Tbps, the NAV-C24S2Q supports comprehensive Layer 2, Layer 3, and Layer 4 switching capabilities, including advanced QoS (Quality of Service) and multicast functions. It efficiently handles voice, video, and data integration, making it suitable for smart campuses, smart cities, and large-scale transportation networks.

Key Features

Ultra-High Bandwidth

- 24× 1/10G SFP+ fiber ports and 2× 40G QSFP uplink ports for high-density core aggregation.
- 2.56Tbps switching capacity and 476Mpps forwarding rate ensure non-blocking, line-speed data transmission.

Comprehensive Security

- Supports IEEE 802.1x, Radius, & Tacacs+ authentication with strict ACL policies.
- Provides hardware-level protection against DoS attacks, TCP SYN/UDP floods, & broadcast storms for network security.

Advanced L3 Routing

- Full support for IPv4/IPv6 routing protocols, including RIP, OSPF, BGP, and Static Routing for complex network interconnections.
- Features VRRP and seamless IPv4-to-IPv6 transition technologies like Tunneling and 6to4.

Carrier-Grade Reliability

- Supports STP/RSTP/MSTP & LACP for millisecond-level link recovery and ring network redundancy.
- Intelligent fan system adjusts speed automatically to reduce noise & energy, backed by an advanced power architecture.

Smart O&M Management

- Complete Ethernet OAM (802.3ah/lag) enables real-time network monitoring and rapid fault localization.
- Supports secure management via Web, CLI, SSH 2.0, and SNMP (V1/V2/V3), plus RMON for traffic statistics.

Featuring 26 optical ports, robust L3 routing protocols, and intelligent security, the NAV-C24S2Q delivers the scalability and stability required for smart campuses and large-scale aggregation layers.

Technical Specification

Category	Specification
Model	NAV-C24S2Q
Interface Characteristics	
Fixed Port	2*40G (QSFP) fiber ports (Data) 24*1/10G SFP+ fiber ports (Data) 1x Console RS232 port (9600,8,N,1) 1x RJ45 port with network management (Data) 1x USB 2.0 port
Optical Fiber Port	10G SFP+ optical fiber interface, default no include optical modules (optional single-mode/ multi-mode, single fiber/ dual fiber optical module. LC)
Optical Cable/ Distance	Multi-mode: 850nm/0-550m, Single-mode: 1310nm/0-40km, 1550nm/0-120km.
Chip Parameter	
Network Management Type	L3
Network Protocol	IEEE802.3u 100Base-TX, IEEE802.3ab 1000Base-T, IEEE802.3z 1000Base-X, IEEE802.3ae 10Gb/s Ethernet, IEEE802.3ba 40Gb/s Ethernet, IEEE802.3x 1
Forwarding Mode	Store and Forward (Full Wire Speed)
Switching Capacity	2.56Tbps (nonblocking)
Forwarding Rate @64byte	476.16Mpps
MAC	128K
Buffer Memory	72M
Jumbo Frame	9K
LED Indicator	Power: PWR (Green), System: SYS (Green)
Power Supply	
Total PWR/ Input Voltage	96W/ (AC100-240V)
Power Consumption	Standby <50W, Full Load <90W
Power Supply Type	Built-in power supply, AC100~240V 50-60Hz, 1.0A
Power Expansion Slot	Default built-in single power supply support dual power supply expansion
Physical Parameter	
Operation Temp/ Humidity	-20°C~+55°C, 5%~90% RH Non condensing
Storage Temp/ Humidity	-40°C~+75°C, 5%~95% RH Non condensing
Dimension (LWH)	440*320*44mm
Net/Gross Weight	<8.0kg / <8.5kg
Installation	Desktop, 1U/19" cabinet
Certification & Warranty	
Lightning Protection	Port lightning protection: 6KV 8/20us, Protection level: IP30
Certification	CCC, CE mark, commercial, CE/LVD EN62368-1, FCC Part 15 Class B, RoHS
Warranty	3 years, lifelong maintenance.
Network Management Feature	
Virtualization and Stacking	VRRP virtualization stack, active/ standby switching
Port Properties	802.3X flow control, Backpressure flow control in half-duplex mode IEEE 802.3az Green Ethernet, Limit the rate of ingress/egress ports Storm suppression based on bandwidth adjustment and storm filtering Storm suppression for unknown unicast, multicast, and broadcast types
VLAN	4K VLAN, Port-based VLAN, 802.1Q standard VLAN

L2 Ring Network Protocol	Standard ERPS ring network Port Fast, Uplink Fast, Backbone Fast BPDU Guard, BPDU Filter, Root Guard Spanning tree ring network: 802.1d STP, 802.1w RSTP, 802.1s MSTP
IPv4	Inter vlan multiple IP, Inter-area route filtering Support multi-domain interoperability Routing COST settings, Plain text/MD5 encryption RIP v1/v2, OSPF, BGP, VRRP, VRRP preemption, BFD and Track Active/standby switchover, Default routing, Policy routing, Static routing
IPv6	IPv6 static routing, RIPng, OSPFv3 IPv6-Ping, IPv6-Tracert, IPv6-Telnet, IPv6-TFTP, IPv6-ICMP
MAC	Static configuration and dynamic learning of MAC addresses View and clear MAC addresses, Automatic update, two-way learning Configurable MAC address aging time, MAC address filtering function
Multicast	IGMP Proxy, IGMP V1/2(Snooping)
DHCP	DHCP Snooping, DHCP Relay, DHCP Server
Mirroring	CPU image Port mirroring for incoming port packets Mirroring of multiple pairs of 1 destination port Perform port mirroring on outgoing port packets
ACL/QOS	ACL verification of source port Access list based on source mac ACL verification of destination port Access lists that support time periods SP, WRR, SP+WRR scheduling mode ACL verification of source IP address Mac access list based on ethernet type ACL verification of destination IP address 8 queues, 802.1P/DSCP priority mapping DSCP priority remarking, IP access list-standard ACL 802.1P(CoS) priority marking, Qos based on DSCP/COS
Security	ARP Aing/Bnding/Proxy, Radius certification Port security, Port-based MAC address filtering Port isolation, MAC address-based authentication Loop detection (private), Single-group port isolation 802.1X remote authentication, IP source protection Storm suppression for multicast/unknown unicast/broadcast Port-based 802.1X local authentication, Multiple port isolation

Management	Restore factory mode, RMON event history SNMP v1/v2/v3, Console, Telnet, SSH 2.0 TFTP file upload and download management NTP, Ping, Tracert, SNMP Alarm/Inform/Traps SNMP (Simple Network Management Protocol) Browser-based WEB management (HTTP, HTTPS) User hierarchical management and password protection Hierarchical alarm, System log (operation log and alarm log)
System	Web browser: Mozilla Firefox 2.5 or higher, Google Chrome V42 or higher, Cat5 and above Ethernet cable TCP/IP, network adapter, and network operating system (such as Microsoft Windows, Linux, Mac OS X) installed on each computer in the network Cat5 and above Ethernet cable

Ordering Information

Model	Description	Power Supply
NAV-C24S2Q	L3 managed Ethernet core routing switch with 24x 1/10G SFP+ fiber ports and 2x 40G QSFP fiber ports. Built-in power supply. Supports 1U/19" cabinet installation.	96W
Note: Optical modules are not included and must be purchased separately.		

Packing List

Content	Qty	Unit
NAV-C24S2Q (26-port 10G core routing switch) 1	1	Set
AC Power Cable 2	1	PC
Mounting Ear 3	1	Set
User Guide 4	1	PC
Warranty Card and Certificate of Conformity 5	1	PC

Optical Module

Product	Model	Description	Unit
1.25G Optical Module	2630	SFP optical module, 1.25G multi-mode dual fiber 850nm, transmission distance: 550m, LC interface. Supports DDM function and hot plugging.	PC
	2632	SFP optical module, 1.25G single-mode dual fiber 1310nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	2612-T	SFP optical module, 1.25G single-mode single fiber TX1310nm/ RX1550nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	2613-R	SFP optical module, 1.25G single-mode single fiber TX1550nm/ RX1310nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	2612-T-SC	SFP optical module, 1.25G single-mode single fiber TX1310nm/ RX1550nm, transmission distance: 20km, SC interface. Supports DDM function and hot plugging.	PC
	2613-R-SC	SFP optical module, 1.25G single-mode single fiber TX1550nm/ RX1310nm, transmission distance: 20km, SC interface. Supports DDM function and hot plugging.	PC
Power Module	2633	1.25G SFP optical module transfers to 10/100/1000M RJ45 port.	PC
10G Optical Module	6630	SFP+ optical module, 10G multi-mode dual fiber 850nm, transmission distance: 300m, LC interface. Supports DDM function and hot plugging.	PC
	7832	SFP+ optical module, 10G single-mode dual fiber 1310nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	7832-33	SFP+ optical module, 10G single-mode single fiber TX1330nm/ RX1270nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC
	7832-27	SFP+ optical module, 10G single-mode single fiber TX1270nm/ RX13300nm, transmission distance: 20km, LC interface. Supports DDM function and hot plugging.	PC